

Multiple Choice

1. **Answer:** D

Options: A) Diode laser; B) Dye laser; C) Free-electron laser; D) Gas laser

Note: Gas lasers operate by exciting free atoms to higher energy levels and allowing them to return to lower energy states, thereby emitting coherent light through the process of stimulated emission.

2. **Answer:** A

Options: A) 4; B) 5; C) 6; D) 20

Note: The angular magnification of a refracting telescope is determined by the ratio of the focal lengths of the objective lens and the eyepiece lens, and since the focal length of the eyepiece is 20 cm and assuming the objective lens has a focal length of 80 cm (100 cm total separation minus 20 cm for the eyepiece), the magnification is $80 \text{ cm} / 20 \text{ cm} = 4$.

3. **Answer:** D

Options: A) 0.25 W; B) 0.5 W; C) 1 W; D) 4 W

Note: The rate of energy dissipation in a resistor is given by the formula $P = V^2/R$, so if the voltage (V) is doubled, the power (P) increases by a factor of four, resulting in a new rate of 4 W.

4. **Answer:** D

Options: A) 1; B) 2; C) 3; D) 5

Note: The correct answer is 5 because the magnetic quantum number m_l can take on values ranging from $-l$ to $+l$, and for $l = 2$, the possible values are $-2, -1, 0, +1$, and $+2$, totaling five distinct values.

5. **Answer:** B

Options: A) Constant temperature; B) Constant volume; C) Constant pressure; D) Adiabatic

Note: In a constant volume process, no work is done on or by the system, so the change in internal energy of the ideal gas is equal to the heat added, as expressed by the first law of thermodynamics ($U = Q - W$, where $W = 0$).

True / False

6. **Answer:** True

Options: A) True; B) False

Note: In a superconductor, Cooper pairs form due to the attractive interaction mediated by phonons, which are quantized vibrations of the ionic lattice, leading to an effective attraction between electrons that enables superconductivity.

7. **Answer:** True

Options: A) True; B) False

Note: The statement is true because with a quantum efficiency of 0.1, on average, 10 out of 100 photons will be detected, and the rms deviation, calculated using the Poisson distribution for the detected photons, is approximately the square root of the average, which is about 3.

8. **Answer:** False

Options: A) True; B) False

Note: The statement is false because the transition from $n = 4, l = 1$ to $n = 3, l = 2$ violates the selection rule that states the orbital angular momentum quantum number (l) can change by at most ± 1 during a transition.

9. **Answer:** True

Options: A) True; B) False

Note: The speed of light in a medium is calculated using the formula $v = \frac{c}{\sqrt{\epsilon_r}}$, where c is the speed of light in a vacuum, and ϵ_r is the dielectric constant; for a dielectric constant of 4.0, this results in a speed of approximately 1.5×10^8 m/s.

10. **Answer:** False

Options: A) True; B) False

Note: The statement is false because the Sun's energy is primarily produced through a series of fusion reactions known as the proton-proton chain, where multiple steps convert hydrogen into helium rather than a direct fusion of two hydrogen atoms into one helium atom.

Long Form Response

11. **Answer:** Fermat's principle of ray optics asserts that light travels along the path that requires the least time, which serves as the foundation for deriving both Snell's law of refraction and the law of reflection. To derive Snell's law, consider the scenario of light passing from one medium to another, where the time taken for light to travel can be expressed as a function of the angles of incidence and refraction relative to the normal. By applying calculus and optimizing this time function, one can derive the relationship between the sine of the angles, leading to Snell's law. Similarly, for the law of reflection, the principle is applied by observing that the angle of incidence equals the angle of reflection, as both angles correspond to paths that require the least time when light reflects off a surface. Thus, Fermat's principle not only unifies these laws but also illustrates the intrinsic nature of light's behavior in varying media.
Note: correct because it correctly identifies that Fermat's principle, which posits that light travels along the quickest path, can be mathematically formulated to derive both Snell's law of refraction and the law of reflection by optimizing the time taken for light to traverse different media and angles.
12. **Answer:** To accelerate a proton from rest to a speed of $0.6c$, we need to consider the relativistic kinetic energy, which is given by the equation $KE = mc^2 - mc^2$, where

(gamma) is the Lorentz factor defined as $\gamma = 1 / \sqrt{1 - v^2/c^2}$. At 0.6 c, γ is approximately 1.25, leading to $KE = (\gamma - 1)mc^2 = 0.25 mc^2$. For a proton with a mass m of approximately $1.67 \times 10^{-27} \text{ kg}$, the work required is about $0.25 \times (1.67 \times 10^{-27} \text{ kg}) \times (3 \times 10^8 \text{ m/s})^2$, resulting in a significant energy requirement. This illustrates the implications of relativistic effects, as the energy needed to reach higher speeds increases dramatically compared to classical mechanics, emphasizing the importance of considering relativistic dynamics in particle acceleration.

Note: The answer correctly uses the relativistic kinetic energy formula and the Lorentz factor to calculate the work required to accelerate a proton to 0.6 c, illustrating the significant energy implications of relativistic effects.

End of Answer Key